
TN1: DERIVING THE FUBINI–STUDY MEASURE AS AN SU(N)-EQUIVARIANT PUSHFORWARD

A PREPRINT

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May 10, 2026

ABSTRACT

Constraint–Surface Dynamics (CSD) models finite-dimensional quantum systems as deterministic Hamiltonian flows on a compact Kähler manifold Σ equipped with Liouville measure μ_L . Observable structure arises via a smooth many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1},$$

which maps ontic microstates to epistemic rays. Previous CSD work assumes that the pushforward measure $\pi_*\mu_L$ is proportional to the Fubini–Study measure μ_{FS} on $\mathbb{CP}^{N-1}$, enabling identification of epistemic volumes with the canonical invariant projective measure.

In this technical note we prove that this proportionality is not an additional postulate but follows from symmetry. Specifically, we show that if Σ is compact, the Liouville measure is invariant under an $SU(N)$ action in the symmetry-compatible quantum-effective sector, and the projection π is smooth, surjective, and $SU(N)$ -equivariant, then

$$\pi_*\mu_L = c\mu_{FS}$$

for a constant $c > 0$. The proof relies only on invariance of Liouville volume and uniqueness of invariant measures on the compact homogeneous space $\mathbb{CP}^{N-1}$. Explicit verification is given for the Hopf fibrations $S^3 \rightarrow \mathbb{CP}^1$ and $S^{2N-1} \rightarrow \mathbb{CP}^{N-1}$.

This result closes the measure bridge in the finite-dimensional symmetry-compatible sector of CSD. It does not by itself supply the general probability rule for arbitrary epistemic preparations, which in the programme is state-conditioned and expressed either by integration against the epistemic density ρ_{ep} or, equivalently, by operator-form probabilities. Within that sector, the note shows that the canonical epistemic reference measure is fixed by Liouville volume and $SU(N)$ symmetry without introducing stochastic primitives.

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1 Introduction

1.1 The Measure-Theoretic Gap

Constraint–Surface Dynamics (CSD) models finite-dimensional quantum systems as deterministic Hamiltonian flows on a compact Kähler manifold Σ equipped with Liouville measure

$$\mu_L = \frac{\omega_\Sigma^d}{d!}, \quad d = \frac{1}{2} \dim_{\mathbb{R}} \Sigma.$$

Observable structure emerges only after applying a smooth many-to-one projection

$$\pi : \Sigma \rightarrow \mathbb{C}P^{N-1},$$

which maps ontic microstates to epistemic rays.

Paper A establishes that long-run outcome frequencies arise as Liouville-volume ratios under measure-preserving dynamics on Σ . Paper B fixes the Born-form probability assignment on projective space from symmetry and operational

consistency. Paper C constructs measurement partitions on $\mathbb{CP}^{N-1}$ using Fubini–Study geometry and shows that their boundaries are μ_{FS} -null. TN0 records the canonical notation and the state-conditioned probability rules used across the programme.

The purpose of the present note is to derive the proportionality

$$\pi_* \mu_L = c \mu_{FS}$$

from symmetry and compactness alone within the symmetry-compatible quantum-effective sector.

1.2 Main Result

We consider the symmetry-compatible quantum-effective sector of CSD. In this sector we assume:

1. Σ is compact and carries a Liouville measure μ_L invariant under a smooth $SU(N)$ action;
2. the projection $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$ is smooth, surjective, and $SU(N)$ -equivariant.

Under these conditions we prove that

$$\pi_* \mu_L = c \mu_{FS}$$

for some constant $c > 0$.

The proof relies on two structural facts: invariance of Liouville volume under the $SU(N)$ action and uniqueness of invariant measures on the compact homogeneous space

$$\mathbb{CP}^{N-1} \cong SU(N)/S(U(N-1) \times U(1)).$$

No assumption is made regarding fibre-volume uniformity. The result is conditional on the existence of the stated symmetry structure and does not derive that structure from deeper physical principles.

1.3 Logical Role Within CSD

The role of the present note is narrow and specific. It supplies a self-contained measure-bridge support result for the finite-dimensional programme: in the symmetry-compatible quantum-effective sector, the pushforward of Liouville measure is proportional to the Fubini–Study measure on $\mathbb{CP}^{N-1}$.

Combined with:

- deterministic volume typicality on Σ (Paper A),
- the Born-form probability assignment fixed by symmetry and operational consistency (Paper B),
- and context-dependent measurement regions on $\mathbb{CP}^{N-1}$ together with the finite-dimensional operational dictionary (Paper C, TN4, TN0),

this identifies the canonical epistemic reference measure used in the finite-dimensional sector.

Appendix A provides a self-contained version of the invariant-measure uniqueness argument used by Paper B and by the present note. The note does not solve the upstream problem of why physical ontic dynamics enter the symmetry-compatible quantum-effective sector. In the programme dependency map, that remains tied to the conditional status of A4b and the open physical grounding of A5.

2 Setup and Assumptions

This section formalizes the geometric and symmetry assumptions required for the pushforward density theorem. No additional ontic structure beyond that already specified in CSD is introduced.

2.1 The Ontic Manifold and Liouville Measure

Let $(\Sigma, \omega_\Sigma, J_\Sigma, g_\Sigma)$ be a compact connected Kähler manifold of real dimension $2d$. The Liouville volume form is

$$\Omega_\Sigma = \frac{\omega_\Sigma^d}{d!},$$

with associated finite Borel measure

$$\mu_L(U) = \int_U \Omega_\Sigma, \quad U \subset \Sigma \text{ measurable.}$$

Compactness ensures $\mu_L(\Sigma) < \infty$.

We restrict attention to the symmetry-compatible quantum-effective sector of CSD, in which Σ carries a smooth $SU(N)$ action

$$\Phi : SU(N) \times \Sigma \rightarrow \Sigma$$

such that

1. $\Phi_g^* \omega_\Sigma = \omega_\Sigma$ for all $g \in SU(N)$,
2. the action preserves the complex structure J_Σ .

It follows that the Liouville measure is $SU(N)$ -invariant:

$$\mu_L(\Phi_g(U)) = \mu_L(U) \quad \forall g \in SU(N).$$

2.2 The Projection Map

Let

$$\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$$

be a smooth surjective map. Since smooth maps between manifolds are Borel measurable, π is measurable.

The pushforward measure on $\mathbb{CP}^{N-1}$ is defined by

$$(\pi_* \mu_L)(A) = \mu_L(\pi^{-1}(A)), \quad A \subset \mathbb{CP}^{N-1} \text{ Borel.}$$

Because μ_L is finite and π is measurable, $\pi_* \mu_L$ is a finite Borel measure on $\mathbb{CP}^{N-1}$.

No assumption regarding fibre-volume uniformity or bundle structure is used in the main theorem.

2.3 $SU(N)$ -Equivariance

The central structural assumption is equivariance of the projection:

$$\pi(\Phi_g(x)) = g \cdot \pi(x), \quad \forall g \in SU(N), x \in \Sigma,$$

where $g \cdot [\psi] = [g\psi]$ denotes the standard $SU(N)$ action on $\mathbb{CP}^{N-1}$.

Equivariance ensures compatibility between ontic symmetry on Σ and epistemic symmetry on projective space.

Lemma 1 (Pushforward invariance). *If μ_L is $SU(N)$ -invariant and π is $SU(N)$ -equivariant, then the pushforward measure $\pi_* \mu_L$ is $SU(N)$ -invariant on $\mathbb{CP}^{N-1}$.*

Proof. For any measurable set $A \subset \mathbb{CP}^{N-1}$,

$$(\pi_* \mu_L)(gA) = \mu_L(\pi^{-1}(gA)).$$

Equivariance gives

$$\pi^{-1}(gA) = \Phi_g(\pi^{-1}(A)).$$

Using invariance of μ_L ,

$$\mu_L(\Phi_g(\pi^{-1}(A))) = \mu_L(\pi^{-1}(A)).$$

Therefore

$$(\pi_* \mu_L)(gA) = (\pi_* \mu_L)(A).$$

□

□

2.4 Reduction of the Problem

The complex projective space admits the homogeneous-space representation

$$\mathbb{CP}^{N-1} \cong SU(N)/S(U(N-1) \times U(1)),$$

and carries the $SU(N)$ -invariant Fubini–Study volume form

$$\Omega_{FS} = \frac{\omega_{FS}^{N-1}}{(N-1)!},$$

with associated Borel measure μ_{FS} .

Throughout this note, μ_{FS} denotes the un-normalised Borel measure associated to the Fubini–Study volume form Ω_{FS} . When a normalised probability measure is intended, it is written explicitly as μ_{FS}^{norm} .

Since $\pi_*\mu_L$ is finite and $SU(N)$ -invariant, the uniqueness theorem for invariant measures on compact homogeneous spaces, recalled in Appendix A, implies

$$\pi_*\mu_L = c \mu_{FS}$$

for some constant $c > 0$.

3 The Pushforward Density Theorem

We now state and prove the central result of this note. The argument relies only on compactness of Σ , $SU(N)$ -invariance of the Liouville measure, equivariance of the projection, and uniqueness of invariant measures on $\mathbb{CP}^{N-1}$ (Appendix A).

3.1 Statement of the Theorem

Theorem 1 (Pushforward Density Theorem). *Let (Σ, ω_Σ) be a compact connected Kähler manifold with Liouville measure*

$$\mu_L = \frac{\omega_\Sigma^d}{d!}, \quad d = \frac{1}{2} \dim_{\mathbb{R}} \Sigma.$$

Assume that:

1. μ_L is invariant under a smooth $SU(N)$ action on Σ ;
2. $\pi : \Sigma \rightarrow \mathbb{CP}^{N-1}$ is smooth, surjective, and $SU(N)$ -equivariant;
3. $\mu_L(\Sigma) > 0$.

Then there exists a constant $c > 0$ such that

$$\pi_*\mu_L = c \mu_{FS},$$

where μ_{FS} denotes the Fubini–Study measure on $\mathbb{CP}^{N-1}$.

3.2 Proof

Step 1: Finiteness: Since Σ is compact,

$$\mu_L(\Sigma) < \infty.$$

Therefore

$$\pi_*\mu_L(\mathbb{CP}^{N-1}) = \mu_L(\Sigma),$$

so $\pi_*\mu_L$ is a finite Borel measure.

Step 2: Invariance: By Lemma 1, equivariance of π and invariance of μ_L imply

$$\pi_*\mu_L(gA) = \pi_*\mu_L(A) \quad \forall g \in SU(N),$$

for every measurable set $A \subset \mathbb{CP}^{N-1}$. Thus $\pi_*\mu_L$ is $SU(N)$ -invariant.

Step 3: Uniqueness of Invariant Measure: $\mathbb{CP}^{N-1}$ is a compact homogeneous space

$$\mathbb{CP}^{N-1} \cong SU(N)/S(U(N-1) \times U(1)).$$

By Appendix A, any finite $SU(N)$ -invariant Borel measure on $\mathbb{CP}^{N-1}$ is proportional to the Fubini–Study measure.

Step 4: Determination of the Constant: Integrating both sides over $\mathbb{C}P^{N-1}$ gives

$$\mu_L(\Sigma) = c \mu_{FS}(\mathbb{C}P^{N-1}),$$

so

$$c = \frac{\mu_L(\Sigma)}{\mu_{FS}(\mathbb{C}P^{N-1})}.$$

Since $\mu_L(\Sigma) > 0$ by hypothesis and $\mu_{FS}(\mathbb{C}P^{N-1}) > 0$, it follows that $c > 0$. □

3.3 Constancy of the Density

Since $\pi_* \mu_L = c \mu_{FS}$ with $c > 0$, we have $\pi_* \mu_L \ll \mu_{FS}$ and the Radon–Nikodym derivative exists. It satisfies

$$\frac{d(\pi_* \mu_L)}{d\mu_{FS}} = c \quad \text{almost everywhere.}$$

The density is therefore constant across $\mathbb{C}P^{N-1}$.

Importantly, this constancy is not imposed through uniform fibre geometry. It follows entirely from compactness and global $SU(N)$ symmetry. Any position-dependent density would violate invariance under the transitive group action.

This completes the proof of the pushforward density theorem.

4 Worked Examples

We now verify Theorem 3.1 explicitly in canonical geometries. These examples illustrate how constant pushforward density arises from symmetry alone and does not require fine-tuning of fibre geometry.

4.1 The Qubit Case: $S^3 \rightarrow \mathbb{C}P^1$

Let

$$\Sigma = S^3 \subset \mathbb{C}^2$$

with the Liouville measure induced from the standard symplectic structure on $\mathbb{C}^2$. The projection

$$\pi : S^3 \rightarrow \mathbb{C}P^1$$

is the Hopf fibration

$$\pi(z_1, z_2) = [z_1 : z_2].$$

Symmetry structure:

- $SU(2)$ acts transitively on S^3 .
- $SU(2)$ acts transitively on $\mathbb{C}P^1$.
- The Hopf projection is $SU(2)$ -equivariant.
- Fibres are circles S^1 .

The standard invariant volume measure on S^3 is $SU(2)$ -invariant. The Fubini–Study measure on $\mathbb{C}P^1$ coincides with the normalized area measure on S^2 under the identification

$$\mathbb{C}P^1 \cong S^2.$$

Volume decomposition: In standard Hopf coordinates

$$z_1 = \cos\left(\frac{\theta}{2}\right) e^{i\alpha}, \quad z_2 = \sin\left(\frac{\theta}{2}\right) e^{i\beta},$$

with fibre coordinate $\chi = \alpha + \beta$ and base azimuth $\varphi = \beta - \alpha$, the standard volume form on S^3 factors as an S^1 fibre factor times the pullback of the Fubini–Study area form on $\mathbb{C}P^1 \cong S^2$. Equivalently,

$$d\mu_{S^3} = d\chi \wedge \pi^*(d\mu_{FS}),$$

up to the chosen normalisation conventions. Integrating over the fibre therefore yields

$$\pi_* \mu_{S^3} = C_1 \mu_{FS}$$

for a positive constant C_1 . If the fibre coordinate is normalised so that $\int_{S^1} d\chi = 2\pi$, then $C_1 = 2\pi$.

4.2 General Hopf Fibration: $S^{2N-1} \rightarrow \mathbb{C}P^{N-1}$

Let

$$\Sigma = S^{2N-1} \subset \mathbb{C}^N$$

with its standard symplectic structure and standard invariant volume measure. The projection

$$\pi : S^{2N-1} \rightarrow \mathbb{C}P^{N-1}$$

is again the Hopf fibration.

Symmetry:

- $SU(N)$ acts transitively on S^{2N-1} .
- $SU(N)$ acts transitively on $\mathbb{C}P^{N-1}$.
- The projection is $SU(N)$ -equivariant.
- Fibres are S^1 .

Measure factorization: In the standard Hopf construction, S^{2N-1} is a principal S^1 bundle over $\mathbb{C}P^{N-1}$. The invariant volume form on S^{2N-1} splits into an S^1 fibre factor and the pullback of the Fubini–Study volume form on $\mathbb{C}P^{N-1}$:

$$d\mu_{S^{2N-1}} = d\chi \wedge \pi^* \Omega_{FS},$$

again up to the chosen normalisation conventions. Therefore

$$\pi_* \mu_{S^{2N-1}} = C_N \mu_{FS}$$

for a positive constant C_N . Under the standard fibre normalisation $\int_{S^1} d\chi = 2\pi$, one has $C_N = 2\pi$. The key point is structural: the pushforward is constant because the construction is $SU(N)$ -equivariant and transitive, not because one separately postulates fibre-wise uniformity.

4.3 Structural Lesson

These examples illustrate the mechanism behind Theorem 3.1:

1. Global $SU(N)$ symmetry enforces invariance of the pushforward measure.
2. Transitivity of the group action forbids position-dependent densities.
3. Constancy of the Radon–Nikodym derivative follows from symmetry alone.

The Hopf fibrations provide explicit realizations, but the theorem does not depend on spherical geometry or principal bundle structure. Compactness and equivariant symmetry are sufficient.

This completes the verification of the pushforward density theorem in canonical examples.

5 Consequences for Epistemic Volume and Measurement Partitions

The pushforward density theorem establishes that, in the quantum-effective sector of CSD,

$$\pi_* \mu_L = c \mu_{FS}.$$

After normalising Liouville measure on Σ ,

$$\tilde{\mu}_L = \frac{\mu_L}{\mu_L(\Sigma)},$$

the induced probability measure on projective space is exactly the normalised Fubini–Study measure:

$$\pi_* \tilde{\mu}_L = \frac{\mu_{FS}}{\mu_{FS}(\mathbb{C}P^{N-1})}.$$

5.1 Measurement Outcome Regions

In Paper C and TN0, a measurement context M determines a measurable partition

$$\{\Omega_i^{\text{ep}}(M)\}_{i \in I} \subset \mathbb{CP}^{N-1},$$

with μ_{FS} -null boundaries. The present note identifies the canonical reference measure on projective space in the symmetry-compatible quantum-effective sector:

$$\pi_* \mu_L = c \mu_{FS}.$$

This identification does not by itself make probabilities state-independent. For a general epistemic preparation represented by a probability measure $\nu_{\text{ep}} \ll \mu_{FS}$ with Radon–Nikodym density ρ_{ep} , the probability of outcome i is

$$P(i \mid \nu_{\text{ep}}, M) = \int_{\Omega_i^{\text{ep}}(M)} \rho_{\text{ep}} d\mu_{FS}.$$

Equivalently, if the preparation is represented in operator form by a density operator $\hat{\rho}$, then for a projective measurement with outcome projectors $\{\Pi_i\}$ one has

$$P(i \mid \hat{\rho}, M) = \text{Tr}(\hat{\rho} \Pi_i).$$

The contribution of the present note is therefore not the derivation of a general probability rule from first principles, but the identification of μ_{FS} as the canonical invariant reference measure entering these state-conditioned constructions.

Only in the special case of the uniform epistemic state does one obtain a state-independent volume ratio:

$$P(i \mid M, \rho_{\text{ep}} = \text{const.}) = \frac{\mu_{FS}(\Omega_i^{\text{ep}}(M))}{\mu_{FS}(\mathbb{CP}^{N-1})}.$$

5.2 Prepared States and Localisation

For sharply prepared rays $[\psi] \in \mathbb{CP}^{N-1}$, TN0 adopts two equivalent idealisations.

First, one may use the operator form

$$\hat{\rho}_{[\psi]} := |\psi\rangle\langle\psi|.$$

Then for a projective measurement with projectors $\{\Pi_i\}$,

$$P(i \mid [\psi], M) = \text{Tr}(\hat{\rho}_{[\psi]} \Pi_i) = \langle\psi \mid \Pi_i \mid \psi\rangle.$$

Second, one may represent sharp preparation by a family of absolutely continuous epistemic measures concentrated near $[\psi]$, for example by restriction and renormalisation to Fubini–Study balls $B_r([\psi])$, and then pass to the localisation limit. Under the regularity assumptions recorded in TN0, one has the differentiation-type expression

$$P(i \mid [\psi], M) = \lim_{r \rightarrow 0} \frac{\mu_{FS}(\Omega_i^{\text{ep}}(M) \cap B_r([\psi]))}{\mu_{FS}(B_r([\psi]))}.$$

For rank-one projective measurements with eigenbasis $\{|\psi_i\rangle\}$, this reduces, by the standard operator rule, to

$$P(i \mid [\psi], M) = |\langle\psi_i, \psi\rangle|^2.$$

The present note does not prove this reduction independently. Its role is narrower: it identifies μ_{FS} as the correct invariant reference measure entering the localisation expression.

5.3 Measure-Theoretic Closure of the Finite-Dimensional Sector

The finite-dimensional sector of CSD now rests on:

- deterministic Hamiltonian flow preserving Liouville measure on Σ ,
- $SU(N)$ -equivariant projection to $\mathbb{CP}^{N-1}$,
- uniqueness of invariant measure on a compact homogeneous space,

- and geometric measurement partitions in projective space.

Within the finite-dimensional symmetry-compatible sector, the measure-theoretic bridge is therefore closed in the following precise sense: the canonical epistemic reference measure is fixed to be μ_{FS} up to normalisation. The full operational probability rule remains state-conditioned, as recorded in TN0, and is supplied in the wider programme by the operator-form and measurement-region constructions rather than by TN1 alone.

This completes the measure-theoretic closure required for subsequent technical notes.

6 Discussion and Limitations

6.1 Scope of the Result

The pushforward density theorem establishes that, within the quantum-effective sector of CSD,

$$\pi_* \mu_L = c \mu_{FS}$$

whenever Liouville measure on Σ is $SU(N)$ -invariant and the projection π is $SU(N)$ -equivariant.

The result is therefore conditional on symmetry. It does not assert that all ontic dynamics on Σ admit such an action. Rather, it applies to those sectors in which the projected dynamics reproduce the standard unitary symmetry of finite-dimensional quantum mechanics.

6.2 Dependence on $SU(N)$ Equivariance

The constancy of the pushforward density follows entirely from global $SU(N)$ symmetry and compactness. If either

- the Liouville measure were not invariant under the $SU(N)$ action, or
- the projection π failed to be $SU(N)$ -equivariant,

then position-dependent densities on $\mathbb{CP}^{N-1}$ would generally be permitted. The proportionality to μ_{FS} is therefore not generic but symmetry-enforced.

In CSD, the required symmetry is assumed in the symmetry-compatible quantum-effective sector, where projected Hamiltonian flows preserve transition probabilities and unitary covariance.

At the programme level, this means that TN1 closes the measure bridge only conditionally: it identifies the pushforward measure in the symmetry-compatible sector, but does not derive from first principles why physical Hamiltonians generically satisfy the upstream quantum-effective or approximate-projectability requirement.

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6.3 Finite-Dimensional Restriction

A separate dimension remark is worth making explicit. The pushforward density theorem itself is dimension-independent within the finite-dimensional setting: it applies for every finite $N \geq 1$ for which the stated compactness and $SU(N)$ -equivariance assumptions hold. The familiar restriction to $N \geq 3$ arises only later, in the Born-weight layer, where Gleason-class theorems are invoked to fix the quadratic probability assignment. That restriction is not a limitation of the measure bridge established here.

The argument relies crucially on compactness of both Σ and $\mathbb{CP}^{N-1}$. In infinite-dimensional Hilbert spaces:

- projective space is non-compact,
- Haar-type uniqueness theorems do not directly apply,
- measure-theoretic subtleties arise.

The present result therefore establishes measure closure only for finite-dimensional systems. Extension to field-theoretic or continuum limits requires separate analysis.

¹Here “measure bridge” refers to the identification of the pushed-forward ontic Liouville measure with the Fubini–Study measure on $\mathbb{CP}^{N-1}$ in the symmetry-compatible sector. The “quantum-effective or approximate-projectability requirement” refers to the condition that ontic Hamiltonian dynamics induces a well-defined effective projected dynamics on $\mathbb{CP}^{N-1}$.

6.4 No Stochastic Assumption

The derivation introduces no probabilistic primitive. What it identifies is the canonical invariant epistemic reference measure

$$\pi_*\mu_L = c\mu_{FS}.$$

General operational probabilities remain state-conditioned. If an epistemic preparation is represented by a probability measure $\nu_{\text{ep}} \ll \mu_{FS}$ with density ρ_{ep} , then

$$P(i \mid \nu_{\text{ep}}, M) = \int_{\Omega_i^{\text{ep}}(M)} \rho_{\text{ep}} d\mu_{FS}.$$

Equivalently, in operator form one has

$$P(i \mid \hat{\rho}, M) = \text{Tr}(\hat{\rho}\Pi_i)$$

for projective measurements.

The role of the present note is to show that μ_{FS} is the unique invariant projective measure compatible with the symmetry assumptions of the finite-dimensional quantum-effective sector.

Only in the special case of the uniform epistemic state does this reduce to a state-independent volume ratio over all of $\mathbb{CP}^{N-1}$.

6.5 Logical Role Within the Programme

Together with deterministic volume typicality on Σ , the Born-form probability assignment on projective space, and context-dependent measurement regions in $\mathbb{CP}^{N-1}$, the pushforward density theorem supplies the measure-bridge component of the finite-dimensional CSD backbone.

More precisely, TN1 establishes the conditional claim

$$\pi_*\mu_L = c\mu_{FS}$$

in the symmetry-compatible quantum-effective sector. It therefore supports the sector-conditional measure bridge used in the finite-dimensional reconstruction. It does not close the upstream question of why physical ontic dynamics lie in that sector, which remains tied to the conditional status of the quantum-effective or approximate-projectability requirement.

Remaining open problems include the physical grounding of the sector-selection assumptions, infinite-dimensional extension, and field-theoretic embedding.

A Uniqueness of the $SU(N)$ -Invariant Measure on $\mathbb{CP}^{N-1}$

This appendix justifies the uniqueness statement used in Theorem 3.1: any finite $SU(N)$ -invariant Borel measure on $\mathbb{CP}^{N-1}$ is proportional to the Fubini–Study measure.

A.1 Homogeneous-Space Structure

The complex projective space admits the identification

$$\mathbb{CP}^{N-1} \cong SU(N)/S(U(N-1) \times U(1)).$$

Thus $\mathbb{CP}^{N-1}$ is a compact homogeneous space G/H with $G = SU(N)$ compact.

The group $SU(N)$ carries a unique normalised Haar probability measure μ_G , invariant under left and right translation.

A.2 Induced Invariant Measure

Let

$$p : G \rightarrow G/H$$

denote the canonical projection. The pushforward of Haar measure defines a G -invariant probability measure on G/H :

$$\mu_{G/H} := p_*\mu_G.$$

This measure satisfies

$$\mu_{G/H}(gA) = \mu_{G/H}(A) \quad \forall g \in G,$$

for all Borel sets $A \subset G/H$.

For $\mathbb{CP}^{N-1}$, this induced invariant measure coincides, up to normalisation, with the Fubini–Study measure associated to the Fubini–Study volume form

$$\Omega_{FS} = \frac{\omega_{FS}^{N-1}}{(N-1)!}.$$

This identification is standard in geometric quantum mechanics and complex projective geometry; see, for example, Bengtsson and Życzkowski, *Geometry of Quantum States*, 2nd ed., Section 4.1, and Ashtekar and Schilling (1999). We denote the associated Borel measure by μ_{FS} and its normalised version by μ_{FS}^{norm} .

A.3 Uniqueness Argument

Let ν be a finite non-zero $SU(N)$ -invariant Borel measure on $\mathbb{CP}^{N-1}$. Normalize:

$$\tilde{\nu} = \frac{\nu}{\nu(\mathbb{CP}^{N-1})}.$$

Then $\tilde{\nu}$ is an $SU(N)$ -invariant probability measure.

Because $\mathbb{CP}^{N-1} \cong G/H$ is a compact homogeneous space with $G = SU(N)$ compact, the G -invariant probability measure on G/H is unique. A standard route is to push forward normalised Haar measure on G along the quotient map $p : G \rightarrow G/H$; uniqueness then follows from the uniqueness of Haar measure and transitivity of the G -action on G/H . Standard references for this construction and uniqueness statement include Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Chapter II, and Knapp, *Lie Groups Beyond an Introduction*, in the discussion of invariant measures on compact homogeneous spaces.

Therefore

$$\tilde{\nu} = \mu_{FS}^{\text{norm}}.$$

Rescaling gives

$$\nu = c \mu_{FS}, \quad c = \frac{\nu(\mathbb{CP}^{N-1})}{\mu_{FS}(\mathbb{CP}^{N-1})}.$$

A.4 Remarks

- Compactness of $SU(N)$ is essential; non-compact groups do not admit finite invariant measures.
- Transitivity of the group action implies that any invariant density must be constant almost everywhere.
- No additional geometric structure beyond homogeneity and compactness is required.

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